10. Let $x$ be a non–zero real number. The equation $x + \frac{1}{x} = k$ has one and only one solution if:

A. $k = 1$

B. $k = 3$

C. $k = -3$

D. $k = -1$

E. $k = 2$

11. The Cartesian equation

$$x^2 + y^2 + 4x = \gamma$$

with $\gamma$ a positive real number, represents:

A. a circumference centred in $(-2,0)$ with radius $\sqrt{\gamma + 4}$

B. a circumference centred in $(0,-4)$ with radius $\sqrt{\gamma + 2}$

C. a circumference centred in $(0,0)$ with radius $\sqrt{\gamma}$

D. a circumference centred in $(-4,0)$ with radius $\sqrt{\gamma + 2}$

E. a circumference centred in $(0,-2)$ with radius $\sqrt{\gamma + 4}$